OpenClaw User Guide

> Integrating the best content from the Orange Book, Datawhale tutorials, APIFox guides, and more
>
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📖 Table of Contents

1. Understanding OpenClaw
2. Quick Start
3. Installation & Deployment
4. Channel Integration
5. Configuration Guide
6. Skills System
7. Memory System
8. Scheduled Tasks
9. Common Commands
10. Troubleshooting

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1. Understanding OpenClaw

1.1 What is OpenClaw?

OpenClaw is an **open-source, self-hosted AI Agent system** that transforms AI from a "chat tool" into a "digital employee capable of autonomously executing tasks."

Dimension	ChatGPT	OpenClaw
Positioning	Q&A Advisor	Digital Employee
Deployment	Cloud SaaS	Local Self-hosted

Dimension	ChatGPT	OpenClaw
Interaction	Web Chat	20+ Chat Channels
Capability	Answer Questions	Execute Tasks
Data	Cloud Storage	Local Control

1.2 Core Capabilities

- ✅ **Multi-channel Connection**: Telegram, WhatsApp, Feishu, DingTalk, Discord, etc.
- ✅ **Persistent Memory**: Remembers your preferences, projects, contacts
- ✅ **Skill Extension**: Infinitely expand capabilities through the Skills system
- ✅ **Proactive Work**: Scheduled tasks, heartbeat checks, automatic reminders
- ✅ **Local Data**: All data stored on your device

1.3 Architecture Overview

```
┌──────────────────────────────────────────────────────────┐
│                    User Chat Channels                    │
│   Telegram / WhatsApp / Feishu / DingTalk / Discord ... │
└──────────────────────┬───────────────────────────────────┘
                       │
                       ▼
┌──────────────────────────────────────────────────────────┐
│                   Gateway Core Service                   │
│  - Message Routing  - Session Management  - Event Processing  - Task
Scheduling │
└──────────────────────┬───────────────────────────────────┘
                       │
                       ▼
┌──────────────────────────────────────────────────────────┐
│                    Agent System                          │
│  main / chat / coding / Custom Agents                    │
└──────────────────────┬───────────────────────────────────┘
                       │
                       ▼
┌──────────────────────────────────────────────────────────┐
│                   Tools & Skills                         │
│  Skills / Plugins / Tools / Memory / Cron               │
└──────────────────────────────────────────────────────────┘
```

2. Quick Start

2.1 System Requirements

Requirement	Description
Node.js	v22.0.0 or higher
Operating System	macOS / Linux / Windows (WSL2)
Memory	Minimum 2GB, Recommended 4GB+
Storage	Minimum 1GB available space

2.2 One-Click Installation

macOS / Linux:

```
curl -fsSL https://openclaw.bot/install.sh | bash
```

Windows (PowerShell):

```
iwr -useb https://openclaw.ai/install.ps1 | iex
```

2.3 Verify Installation

```
# Check version
openclaw --version

# Run diagnostics
openclaw doctor

# Check status
openclaw status
```

2.4 5-Minute Quick Experience

```
# 1. Install
npm i -g openclaw

# 2. Run setup wizard
openclaw onboard

# 3. Start Gateway
openclaw gateway start

# 4. Open dashboard
openclaw dashboard

# 5. Start chatting in your browser!
```

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3. Installation & Deployment

3.1 Installation Methods Comparison

Method	Difficulty	Use Case
One-Click Script	⭐	Recommended for beginners
npm Manual Install	⭐⭐	Familiar with Node.js
Docker	⭐⭐	Containerized deployment
Source Build	⭐⭐⭐⭐	Developers

3.2 Manual Installation (npm)

```
# Check Node.js version
node -v  # Requires >= 22.0.0

# Global installation
npm install -g openclaw@latest

# If encountering sharp module errors
SHARP_IGNORE_GLOBAL_LIBVIPS=1 npm install -g openclaw@latest
```

3.3 Configure PATH (if needed)

If you get `openclaw: command not found`:

```
# Find npm global path
npm prefix -g

# Add to ~/.bashrc or ~/.zshrc
export PATH="$(npm prefix -g)/bin:$PATH"

# Apply changes
source ~/.bashrc
```

3.4 Docker Deployment

```
docker run -d \
  --name openclaw \
  -v ~/openclaw-data:/root/.openclaw \
  -p 18789:18789 \
  openclaw/openclaw:latest
```

3.5 Version Updates

```
# Update to stable version (recommended)
openclaw update --channel stable

# Update to beta version
openclaw update --channel beta

# Update to dev version
openclaw update --channel dev
```

---

4. Channel Integration

4.1 Channel Difficulty Comparison

Tier	Platform	Time	Description
First Tier	Telegram, QQ	5-10 min	Easiest, zero threshold
Second Tier	Discord, Feishu	15-20 min	Well-documented
Third Tier	WhatsApp, Slack, DingTalk	25-40 min	More configuration
Fourth Tier	iMessage, WeChat	1+ hour	Additional requirements

4.2 Telegram Integration (Recommended for Beginners)

Step 1: Create Bot

1. Search for `@BotFather` in Telegram
2. Send `/newbot` command
3. Set bot name (must end with `bot`)
4. Save the returned Bot Token

Step 2: Configure OpenClaw

Edit `~/.openclaw/openclaw.json`:

```
{
  "channels": {
    "telegram": {
      "enabled": true,
      "botToken": "YOUR_BOT_TOKEN",
      "dmPolicy": "pairing",
      "allowFrom": ["tg:YOUR_USER_ID"]
    }
  }
}
```

Step 3: Start and Pair

```
# Restart Gateway
openclaw gateway restart

# Send a message to the bot in Telegram
# You'll receive a pairing verification code
```

4.3 Feishu Integration

```
# Run configuration wizard
openclaw configure --section feishu
```

You need to create an enterprise self-built app in the Feishu Open Platform and obtain App ID and App Secret.

4.4 Multi-Channel Configuration

You can configure multiple channels simultaneously:

```
{
  "channels": {
    "telegram": { "enabled": true, "botToken": "..." },
    "feishu": { "enabled": true, "appId": "...", "appSecret": "..." },
    "discord": { "enabled": true, "botToken": "..." }
  }
}
```

---

5. Configuration Guide

5.1 Configuration File Locations

File	Path	Description
Main Config	`~/.openclaw/openclaw.json`	Gateway, channels, models
Agent Config	`~/.openclaw/agents/*/agent/`	Individual agent configs
Workspace	`~/.openclaw/workspace-*/`	Agent workspaces

5.2 Model Configuration

Edit `~/.openclaw/openclaw.json`:

```
{
  "models": {
```

```
    "mode": "merge",
    "providers": {
      "bailian": {
        "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1",
        "apiKey": "sk-YOUR_API_KEY",
        "api": "openai-completions",
        "models": [
          {
            "id": "qwen3.5-plus",
            "name": "qwen3.5-plus",
            "contextWindow": 1000000,
            "maxTokens": 65536
          }
        ]
      }
    }
  },
  "agents": {
    "defaults": {
      "model": {
        "primary": "bailian/qwen3.5-plus"
      }
    }
  }
}
```

5.3 Recommended Models

Provider	Model	Use Case
Alibaba Cloud	qwen3.5-plus	General tasks, cost-effective
Alibaba Cloud	qwen3-coder-plus	Programming, code generation
Zhipu	glm-5	Optimized for Chinese tasks
Moonshot AI	kimi-k2.5	Long text processing

5.4 Gateway Configuration

```
{
  "gateway": {
    "port": 18789,
```

```
    "mode": "local",
    "bind": "lan",
    "auth": {
      "mode": "token",
      "token": "your-secure-token"
    },
    "controlUi": {
      "allowedOrigins": ["http://192.168.1.100:18789"],
      "allowInsecureAuth": true
    }
  }
}
```

5.5 Authentication Modes

Token Authentication (Recommended for API Integration):

```
{
  "gateway": {
    "auth": {
      "mode": "token",
      "token": "your-token"
    }
  }
}
```

Password Authentication (Recommended for Web UI):

```
{
  "gateway": {
    "auth": {
      "mode": "password"
    }
  }
}
```

---

6. Skills System

6.1 What are Skills?

Skills are OpenClaw's plugin system, enabling AI to:

- Access external APIs (weather, news, stocks)
- Operate local resources (files, cameras)
- Execute specific tasks (GitHub, calendar, email)

6.2 Installing Skills

```
# Install from ClawHub
openclaw skill install <skill-slug>

# Search Skills
openclaw skill search <keyword>

# List installed
openclaw skill list

# Update Skills
openclaw skill update
```

6.3 Popular Skills

Skill	Function	Install Command
weather	Weather forecast	`skill install weather`
github	GitHub operations	`skill install github`
news-aggregator	News aggregation	`skill install news-aggregator`
pinchtab-browser	Browser automation	`skill install pinchtab-browser`

6.4 Creating Custom Skills

```
# Create Skill directory
mkdir -p ~/.openclaw/workspace-main/skills/my-skill

# Create SKILL.md
cat > ~/.openclaw/workspace-main/skills/my-skill/SKILL.md << 'EOF'
# my-skill

## Description
```

```
My custom skill

## Usage
/MyCommand [arguments]
EOF
```

---

7. Memory System

7.1 Memory Types

Type	File	Description
Long-term Memory	`MEMORY.md`	Curated important information
Daily Memory	`memory/YYYY-MM-DD.md`	Raw logs
Identity	`IDENTITY.md`	AI personality settings
User	`USER.md`	User information
Soul	`SOUL.md`	Behavioral guidelines

7.2 Memory File Locations

```
~/.openclaw/workspace-main/
├── MEMORY.md              # Long-term memory
├── IDENTITY.md            # Identity settings
├── USER.md                # User information
├── SOUL.md                # Behavioral guidelines
├── AGENTS.md              # Workspace description
├── TOOLS.md               # Tool notes
├── HEARTBEAT.md           # Heartbeat tasks
└── memory/
    ├── 2026-03-16.md      # Daily memory
    └── 2026-03-15.md
```

7.3 Search Memory

```
# Semantic search
openclaw memory search "project mentioned last time"

# Search by date
openclaw memory search --date 2026-03-15 "meeting"
```

7.4 Memory Maintenance

Recommended to review MEMORY.md regularly (weekly):

- Remove outdated information
- Add new decisions and context
- Organize highlights from daily memory files

---

8. Scheduled Tasks

8.1 Cron Basics

```
# Add scheduled task
openclaw cron add --schedule "0 9 * * *" --message "Good morning! Today's
tasks:"

# List tasks
openclaw cron list

# Run task
openclaw cron run <job-id>

# Remove task
openclaw cron remove <job-id>
```

8.2 Cron Expressions

Expression	Description
`0 9 * * *`	Daily at 9:00
`0 9 * * 1`	Every Monday at 9:00

Expression	Description
`*/30 * * * *`	Every 30 minutes
`0 0 1 * *`	1st of each month at 0:00

8.3 Heartbeat Tasks

Edit `HEARTBEAT.md`:

```
# HEARTBEAT.md

- [ ] Check unread emails
- [ ] View calendar (within 24 hours)
- [ ] Check weather (if going out)
- [ ] Review memory files (weekly)
```

8.4 Practical Cron Examples

Daily News Briefing:

```
openclaw cron add \
  --name "daily-news" \
  --schedule "0 8 * * *" \
  --message "Execute news-aggregator skill, send daily news"
```

Weekly Backup:

```
openclaw cron add \
  --name "weekly-backup" \
  --schedule "0 2 * * 0" \
  --message "Backup workspace and memory directories"
```

---

9. Common Commands

9.1 Gateway Management

```
openclaw gateway start      # Start
openclaw gateway stop       # Stop
openclaw gateway restart    # Restart
openclaw gateway status     # Status
openclaw gateway install    # Install as system service
openclaw gateway uninstall  # Uninstall service
```

9.2 Configuration Management

```
openclaw onboard            # Initial setup wizard
openclaw configure          # Configure channels
openclaw doctor             # Health check
openclaw status             # Check status
openclaw dashboard          # Open web dashboard
openclaw tui                # Terminal interface
```

9.3 Model Management

```
openclaw models status      # Model status
openclaw models list        # List models
openclaw models add         # Add model
```

9.4 Channel Management

```
openclaw channels list      # List channels
openclaw channels add       # Add channel
openclaw channels remove    # Remove channel
```

9.5 Memory Operations

```
openclaw memory search <query>   # Search memory
openclaw memory list             # List memory files
openclaw memory edit             # Edit memory
```

9.6 Session Management

```
openclaw sessions list           # List sessions
openclaw sessions history <key>  # View history
openclaw sessions send           # Send message to session
```

---

10. Troubleshooting

10.1 Common Issues

Issue 1: `openclaw: command not found`

```
# Find npm global path
npm prefix -g

# Add to PATH
export PATH="$(npm prefix -g)/bin:$PATH"

# Add to ~/.bashrc permanently
echo 'export PATH="$(npm prefix -g)/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

Issue 2: Gateway fails to start

```
# Check port usage
lsof -i :18789

# View logs
journalctl --user -u openclaw-gateway.service -n 50

# Restart service
systemctl --user restart openclaw-gateway.service
```

Issue 3: Not receiving Telegram messages

```
# Check if Bot Token is correct
cat ~/.openclaw/openclaw.json | grep botToken

# Check if offset is updated
cat ~/.openclaw/telegram/update-offset-*.json
```

```
# View Gateway logs
journalctl -t gateway-start.sh -n 30 | grep telegram
```

Issue 4: Model API call failures

```
# Check API Key
openclaw models status

# Test connection
curl -H "Authorization: Bearer YOUR_API_KEY" \
  https://api.example.com/v1/models

# Check network
ping api.example.com
```

10.2 Viewing Logs

```
# Gateway logs
journalctl --user -u openclaw-gateway.service -f

# Last 50 entries
journalctl --user -u openclaw-gateway.service -n 50

# Specific time range
journalctl --user -u openclaw-gateway.service \
  --since "2026-03-16 10:00:00" --until "2026-03-16 12:00:00"
```

10.3 Diagnostic Commands

```
# Full diagnostics
openclaw doctor

# Check system
openclaw status

# Check configuration
openclaw config list
```

10.4 Reset & Recovery

```
# Reset configuration (use with caution)
openclaw onboard --reset

# Reinstall Gateway
openclaw gateway uninstall
openclaw gateway install

# Clear cache
rm -rf ~/.openclaw/browser
rm -rf ~/.openclaw/cron
```

Appendix

A. Resource Links

Resource	Link
Official Docs	https://docs.openclaw.ai
GitHub	https://github.com/openclaw/openclaw
ClawHub	https://clawhub.com
Discord	https://discord.com/invite/clawd

B. Version Information

- Document Version: v1.0
- Applicable OpenClaw Version: v2026.3.11+
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C. Contributors

This document integrates content from:

- OpenClaw Orange Book (Hua Shu)
- Datawhale OpenClaw Tutorial
- APIFox OpenClaw User Guide
- OpenClaw Official Documentation

For questions or suggestions, please provide feedback to the OpenClaw community.